

---

# The Neuropsychiatric Inventory:

## Comprehensive assessment of psychopathology in dementia

J.L. Cummings, MD; M. Mega, MD; K. Gray, MD; S. Rosenberg-Thompson, RN, MN;  
D.A. Carusi, BS; and J. Gornbein, DrPH

---

**Article abstract**—We developed a new instrument, the Neuropsychiatric Inventory (NPI), to assess 10 behavioral disturbances occurring in dementia patients: delusions, hallucinations, dysphoria, anxiety, agitation/aggression, euphoria, disinhibition, irritability/lability, apathy, and aberrant motor activity. The NPI uses a screening strategy to minimize administration time, examining and scoring only those behavioral domains with positive responses to screening questions. Both the frequency and the severity of each behavior are determined. Information for the NPI is obtained from a caregiver familiar with the patient's behavior. Studies reported here demonstrate the content and concurrent validity as well as between-rater, test-retest, and internal consistency reliability; the instrument is both valid and reliable. The NPI has the advantages of evaluating a wider range of psychopathology than existing instruments, soliciting information that may distinguish among different etiologies of dementia, differentiating between severity and frequency of behavioral changes, and minimizing administration time.

NEUROLOGY 1994;44:2308-2314

---

Neuropsychiatric disturbances are common manifestations of dementing disorders. Alzheimer's disease (AD) is accompanied by delusions, agitation, anxiety, and personality changes<sup>1-6</sup>; frontotemporal dementias are evidenced by apathy, euphoria, irritability, and disinhibition<sup>7,8</sup>; and vascular dementia produces apathy, irritability, and lability.<sup>7,9,10</sup> Neuropsychiatric disorders may be the presenting manifestations of dementing disorders, appearing before cognitive alterations and heralding the onset of the brain disease.<sup>11-13</sup> Neuropsychiatric manifestations change as the dementias progress, requiring reevaluation and implementation of new interventions in the course of the illness.<sup>14</sup> Neuropsychiatric disturbances add to caregiver burden and are a major cause of institutionalization of dementia patients.<sup>15-17</sup> Some behavioral changes have prognostic significance. Delusions, for example, are associated with more rapid decline in intellectual function in AD.<sup>18-22</sup> The neuropsychiatric aspects of dementia syndromes are often treatable with psychotropic medications.<sup>2,7,23</sup> Thus, neuropsychiatric features of dementias have important diagnostic, prognostic, and management implications.

The assessment and measurement of neuropsychiatric disturbances in dementia present unique challenges. At the time of the interview, patients may not

exhibit, remember, or report their symptoms; some neuropsychiatric disorders, such as delusions and hallucinations, are entirely subjective and if not described by the patient must be inferred from their behavior; the magnitude of behavior changes may be related to either their severity or their frequency; and patients may simultaneously exhibit dementia, depression, and anxiety, making it difficult to distinguish among types of behavioral change. Several tools are available to assess behavior in dementia patients. The principal instruments currently in use include the BEHAVE-AD,<sup>2</sup> the Hamilton Depression Rating Scale (HDRS),<sup>24</sup> the Neurobehavior Rating Scale,<sup>25</sup> the Columbia University Scale for Psychopathology in Alzheimer's Disease,<sup>26</sup> and the Cornell Scale for Depression in Dementia.<sup>27</sup> The BEHAVE-AD<sup>2</sup> is the instrument most widely used for the evaluation of dementia-related behavioral changes. Developed specifically to assess patients with AD, the BEHAVE-AD addresses delusions, hallucinations, activity disturbances, aggressiveness, diurnal rhythm disturbances, affective disturbances, and anxieties and phobias. It does not evaluate other types of behavioral change commonly accompanying AD (eg, apathy or irritability) or occurring in other types of dementia (eg, disinhibition in frontotemporal dementias).

---

From the Departments of Neurology (Drs. Cummings and Mega, and D.A. Carusi) and Psychiatry and Biobehavioral Science (Drs. Cummings and Gray, and S. Rosenberg-Thompson), UCLA School of Medicine, Behavioral Neuroscience Section, Psychiatry Service; West Los Angeles Veterans Affairs Medical Center (Drs. Cummings and Gray); and the Biomathematical Consulting Unit, Department of Biomathematics (Dr. Gornbein), UCLA School of Medicine, Los Angeles, CA.

Supported by the Department of Veterans Affairs, a National Institute on Aging Alzheimer's Disease Core Center grant (AG10123), and the Irving and Helga Cooper Award.

Received February 1, 1994. Accepted in final form June 7, 1994.

Address correspondence and reprint requests to Dr. Jeffrey L. Cummings, Neurobehavior Unit, Psychiatry Service (116AF), West Los Angeles VAMC, 11301 Wilshire Boulevard, Los Angeles, CA 90073.

Depression is frequently associated with dementia. Major depressive episodes are common in vascular dementia and in the subcortical dementias accompanying movement disorders such as Parkinson's disease.<sup>7</sup> Depressive symptoms occur in up to 40% of AD patients.<sup>2,5,9,28</sup> The HDRS<sup>24</sup> is frequently used to assess depression in dementia; it is usually administered with a scripted text.<sup>29</sup> The reliability of the instrument is acceptable.<sup>30</sup> Developed for use in nondemented patients, the HDRS assesses several areas that occur both in depression and in nondepressed demented patients—agitation, weight loss, insomnia, impaired concentration—making it difficult to know whether the scale score reflects depression or dementia. The HDRS also depends on gathering information from demented patients who may not remember their symptoms; the validity of the alternative strategy of asking a spouse about a patient's mood is uncertain.

The Neurobehavior Rating Scale was developed for assessment of behavioral changes following head trauma.<sup>25</sup> It is also used to evaluate behavioral changes in dementia and distinguishes behaviors in AD from those of vascular dementia.<sup>28</sup> The instrument has 28 items, each receiving a scale score of 0 to 6 after exploratory questions have been asked. It thus requires considerable time to administer and cannot be readily used in routine clinical settings. Scores are based on patient observation by the clinician and may underestimate psychopathology not evident during the examination.

A number of other instruments have recently become available that focus on the assessment of behavioral changes in dementia. These include the Columbia University Scale of Psychopathology in Alzheimer's Disease<sup>26</sup> and the Cornell Scale for Depression in Dementia.<sup>27</sup> These tools provide advances over earlier instruments, but they assess a relatively restricted range of behavior and lack differential diagnostic capability. These scales have substantially advanced understanding of the behavioral aspects of dementias and each has an important place in the armamentarium used by clinicians to assess neuropsychiatric disturbances, but development is warranted of new instruments that apply to other behaviors commonly observed in dementia patients.

We developed the Neuropsychiatric Inventory (NPI) to assess a wide range of behaviors encountered in dementia patients, to provide a means of distinguishing frequency and severity of behavioral changes, and to facilitate rapid behavioral assessment through the use of screening questions. We report the content validity, concurrent validity, internal consistency reliability, item independence, between-rater reliability, and test-retest reliability of the NPI. Relationships between NPI scores and age and dementia severity as well as the NPI profiles of 40 normal controls are also described. These studies demonstrate that the NPI is psychometrically secure and can be used to efficiently and comprehensively assess the neuropsychiatric aspects of dementing disorders. This is not a diagnostic study; profiles for different dementias and the ability of the NPI to distinguish among

different types of dementia will be reported later.

**Methods. Content.** The behavioral domains to be assessed were chosen after a comprehensive review of the literature pertaining to behavior occurring in different dementias<sup>1-7</sup> and preliminary studies of behavioral changes in patients with AD, vascular dementia, and frontotemporal dementias.<sup>4,8-10,12,14,28,31-33</sup> Ten behavioral domains are evaluated by the NPI. The domains and a sample question from each are as follows: delusions (Does the patient believe that others are stealing from her or him?); hallucinations (Does the patient talk to people who are not there?); agitation/aggression (Is the patient uncooperative, resistive to help from others?); dysphoria (Does the patient say or act as if she or he is sad or in low spirits?); anxiety (Does the patient say that she or he is worried about planned events?); euphoria (Does the patient find humor in and laugh at things that others do not find funny?); apathy (Does the patient seem less spontaneous and less active than usual?); disinhibition (Does the patient say crude things or make sexual remarks that he or she would not usually say?); irritability/lability (Does the patient have sudden flashes of anger?); and aberrant motor activity (Does the patient pace around the house without apparent purpose?). These domains were chosen because they are commonly affected in dementias and because the items have the potential to distinguish among dementias that produce different types of behavioral disturbances (eg, more disinhibition in frontotemporal dementias, more indifference in AD, more depression in vascular dementia<sup>8,9,12,28</sup>).

**Screening and assessment questions.** One important goal of the NPI is to facilitate behavioral assessment by minimizing the time required to complete the evaluation while providing a comprehensive characterization of abnormal behavior manifested by the patient. This goal was accomplished by developing screening questions that provide an overview of each specific behavioral domain. Information is gathered from caregivers familiar with the patient's behavior. To serve as an informant, the caregiver must have at least daily contact with the patient. Using scripted questions, the caregiver is asked whether the patient's behavior changed after the onset of the dementia and if the altered behavior has been present during the past month or other specific period. If the caregiver indicates that the abnormal behavior was present, the behavioral domain is then explored with either seven or eight subquestions (illustrated above) that provide more detailed information about the specific features of the behavioral disturbance. After the subquestions are answered, the caregiver is asked to rate the severity (1 = mild, 2 = moderate, 3 = severe) and frequency (1 = occasionally, less than once per week; 2 = often, about once per week; 3 = frequently, several times per week but less than every day; 4 = very frequently, once or more per day or continuously) of the behavior. The questions are read aloud by the examiner; defined anchor points for severity and frequency allow caregivers to provide a ratable answer. Using this approach, domains with a negative response to the screening question are not explored, while data regarding the characteristics, severity, and frequency are garnered for each domain with a positive response to the screening question. Caregivers provide the information for the NPI because patients may not remember their behavior when asked, may not exhibit abnormal behavior while with the clinician, and often cannot grasp scaling concepts necessary for determining severity and frequency. Information from observing the patient is not used in scoring the NPI because the examiner is usually not in a position to determine the frequency of behaviors or the range of sever-

ity of behaviors that may be exhibited. If the clinician is aware of information that contradicts the caregiver's report, the information from the NPI is invalid and should be discarded.

After the assessment strategy and behavioral domains were chosen, administration instructions were developed and the NPI was administered to the caregivers of 20 dementia patients to determine the sensitivity of the screening questions. This was done by asking both the screening questions and all the subquestions and determining the number of instances in which a caregiver responded negatively to the screening questions and positively to one of the subquestions.

**Content validity.** The NPI addresses behaviors that have been subjected to relatively little investigation, such as apathy, irritability, disinhibition, and euphoria. Concurrent validity could not be assessed when there was no standard instrument with which the NPI could be compared. To determine the content validity of the NPI, a Delphi panel was developed and asked to rate the scale items. The NPI was submitted to 10 nationally and internationally known experts in geriatric psychiatry, behavioral neurology, or neuropsychology (Delphi panel members are listed in the Acknowledgments). Each panel member rated the screening questions and the subquestion section of each behavioral domain as to whether the essential elements of the behavior were captured. Screening questions and subquestions were rated between 1 (well assessed) and 4 (poorly assessed); operationalized criteria for scoring were provided.

**Concurrent validity.** Concurrent validity was determined by comparing the scores on the relevant subscales of the NPI with the appropriate scales of two instruments currently in widespread use. The NPI, the BEHAVE-AD, and the HDRS were administered to the caregivers of 40 patients with dementia syndromes. The dementias were of a variety of types (AD, vascular dementia, mixed AD-vascular dementia, frontotemporal dementias, and miscellaneous); cognitive impairment had a broad spectrum of severity; and the patients exhibited a wide range of behavioral disorders of varying severity. The delusions, hallucinations, dysphoria, anxiety, agitation/aggression, and aberrant motor behavior subscales of the NPI were compared with the Delusions, Hallucinations, Affective disturbances, Anxiety and phobias, Aggressiveness, and Activity disturbances subscales of the BEHAVE-AD; the Dysphoria subscale of the NPI was compared with the depression score of the HDRS.

**Between-rater reliability.** Between-rater reliability was determined by having two raters score the NPI responses of a caregiver. Raters were familiarized with the instrument through the administration instructions and by observing several interviews conducted by experts using the NPI. Two raters were present during the same interview, which was conducted by one of the raters. The two raters were not the same in each interview. Raters participating in this aspect of NPI assessment included a behavioral neurologist, four behavioral neurology fellows, a geriatric psychiatry fellow, and a premedical student. Each rater was blind to the rating chosen by the other, and raters did not confer on responses before giving the ratings to a third individual who recorded the data. Between-rater reliability interviews were conducted with 45 caregivers.

**Test-retest reliability.** Test-retest reliability was determined by conducting a second NPI interview with caregivers within 3 weeks of an initial interview. The second interview was conducted over the telephone by a clinician not involved in the first interview. Twenty caregivers participated in the test-retest reliability assessment of the NPI.

**Normal control subjects.** To determine the rate of behav-

iors assessed by the NPI in the nondemented population, 40 spouses of nondemented individuals were interviewed using the NPI. The subjects were residents of a Southern California retirement community (Leisure World). Each member of a couple was administered the Mini-Mental State Examination<sup>34</sup> (MMSE), and the spouse was asked if the partner exhibited any signs of dementia or memory loss. If the partner's MMSE score was 25 or above and the spouse denied any concern about the partner's intellectual function, then the spouse was interviewed using the NPI, providing data pertaining to the partner. As with patient caregivers, the interviewees were asked about behavior that differed from the partner's usual behavior and had occurred within the preceding 4 weeks.

**Statistical analyses.** A variety of statistical procedures were used to assess the psychometric properties of the NPI. Concurrent validity was determined by calculating the Spearman correlation coefficients between the NPI subscale scores and the subscales of the BEHAVE-AD or HDRS. Similar methods were used to explore the relationships between NPI subscale scores and age and MMSE scores. Cronbach's coefficient alpha<sup>35</sup> was determined to assess internal consistency reliability, and Pearson's correlations were calculated for the item-independence study. Percent-agreement analyses were used for between-rater reliability, and correlation coefficients were generated for test-retest reliability calculations. When a synoptic score was superior for clinical or analytic purposes, the product of the frequency and the severity (severity of the behavior multiplied by how often it occurred) was used as an overall measure of the magnitude of the behavioral disturbance.

**Results. Patients, caregivers, and controls.** Caregivers providing behavioral information in this study were all family members of outpatients attending a university or a Veterans Affairs dementia clinic or of subjects participating in a clinical trials program. Forty caregivers participated in the concurrent validity study. The patients on whom information was collected were 22 men and 18 women aged 56 to 90 years (mean, 75.7). All patients were assessed with the MMSE<sup>34</sup> and had scores ranging from 0 to 29 (mean, 19.2). Twenty of these patients had AD, nine had vascular dementia, and 11 had other types of dementing disorders. Patients with AD met the criteria of the National Institute of Neurological and Communicative Diseases and Stroke/Alzheimer's Disease and Related Disorders Association<sup>36</sup> for probable AD; patients with vascular dementia met criteria of the National Institute of Neurological Disease and Stroke<sup>37</sup> for probable vascular dementia.

Forty-five NPI assessments were included in the interrater reliability study. Participants were caregivers of outpatients in a dementia clinic or of patients who were on stable doses of medication in a clinical trials program. Their patients were 26 men and 19 women; patient MMSE scores ranged from 1 to 29 (mean, 17.4). Forty-two of the patients had AD, one had vascular dementia, and two had other types of dementing illnesses. A subgroup of 20 of the caregivers participated in the test-retest reliability study. The patients of all these caregivers had AD. Another subgroup of 20 caregivers of patients with a variety of dementias provided the data used in determining the

sensitivity of the screening questions.

Forty normal control subjects participated in the study. MMSE scores ranged from 25 to 30 (mean, 28.4). There were 20 men and 20 women.

**Screening question sensitivity.** The principal requirement when using the screening question strategy is to show that the screen detects nearly all the psychopathology that would be found if all the questions on the scale were asked. Among the 20 patients evaluated in this portion of the NPI assessment, there were nine false-negative responses (20 patients  $\times$  10 behaviors per patient = 200 behaviors; nine "misses" = 4.5%). A false-negative rate of under 5% was deemed acceptable.

False-positive responses also occurred where caregivers responded positively to the screen but negatively to all subquestions. This occurred primarily in the dysphoria section (four of nine false-positive responses) of the NPI. Caregivers responded affirmatively to screening questions using the word "depression" but applied the word in a nonclinical sense to describe behavior determined by the clinician to be unrelated to depressive symptomatology.

**Content validity.** The results of the Delphi panel assessment of the content validity of the NPI are shown in table 1. A score of 1 indicated that the domain was well assessed and a score of 4 indicated that the domain was poorly assessed by the NPI screening questions and subquestions. Each group of questions except those assessing "troublesome behavior" received scores of less than 2, indicating high content validity. "Troublesome behavior" was reformulated as "aberrant motor behavior" in accordance with the recommendations of the Delphi panel.

**Item and summary scores.** Table 2 shows the NPI subscale scores for the 40 subjects in the concurrent validity study. These data show that a wide range of psychopathology was present in the study subjects

although the mean scale scores tended to be low. Apathy was the behavior most commonly recorded; agitation/aggression, dysphoria, anxiety, irritability/lability, and aberrant motor behavior were also frequently described by caregivers. Euphoria was the behavior least often observed and with the most restricted range of expression; this reflects the small number of patients with euphoric behavior included in the sample.

**Concurrent validity.** Concurrent validity was determined by comparing the NPI subscales with the appropriate subscales of the BEHAVE-AD and the total score of the HDRS. Table 3 presents the correlations of the BEHAVE-AD and the HDRS with the NPI sub-

**Table 1. Delphi panel member responses regarding the content validity of the Neuropsychiatric Inventory (NPI)**

| Behavior              | Panel member scores       |                     |
|-----------------------|---------------------------|---------------------|
|                       | Screening question (mean) | Subquestions (mean) |
| Delusions             | 1-2 (1.5)                 | 1-2 (1.4)           |
| Hallucinations        | 1-2 (1.3)                 | 1-3 (1.3)           |
| Dysphoria             | 1-2 (1.2)                 | 1-2 (1.3)           |
| Anxiety               | 1-2 (1.5)                 | 1-2 (1.5)           |
| Euphoria              | 1-2 (1.6)                 | 1-2 (1.3)           |
| Aggression            | 1-2 (1.3)                 | 1-2 (1.2)           |
| Apathy                | 1-3 (1.3)                 | 1-2 (1.4)           |
| Irritability          | 1-2 (1.3)                 | 1-2 (1.7)           |
| Disinhibition         | 1-2 (1.2)                 | 1-2 (1.2)           |
| Troublesome behavior* | 1-3 (2.2)                 | 1-3 (2.0)           |

\* This section was reformulated in accord with the recommendations of the Delphi panel members and appears as "aberrant motor behavior" on the current NPI and in the remainder of the analyses.

**Table 2. Scores of the Neuropsychiatric Inventory subscales (frequency and severity) derived from the concurrent validity study, interrater reliability test results, and results of the study of test-retest reliability**

| Behavior                | Frequency mean (SD; range*) | Severity mean (SD; range) | Between-rater agreement |              | Test-retest reliability   |                          |
|-------------------------|-----------------------------|---------------------------|-------------------------|--------------|---------------------------|--------------------------|
|                         |                             |                           | Frequency (%)           | Severity (%) | Frequency Correlation (p) | Severity Correlation (p) |
| Delusions               | 0.48 (1.11; 0-4)            | 0.28 (0.60; 0-2)          | 100                     | 100          | 0.92 (0.0001)             | 0.80 (0.0001)            |
| Hallucinations          | 0.25 (0.71; 0-3)            | 0.15 (0.36; 0-1)          | 97.9                    | 97.9         | 0.98 (0.0001)             | 1.0 (0.0)                |
| Agitation/Aggression    | 1.28 (1.52; 0-4)            | 0.70 (0.82; 0-3)          | 95.7                    | 95.7         | 0.51 (0.02)               | 0.75 (0.0002)            |
| Dysphoria               | 1.35 (1.56; 0-4)            | 0.70 (0.82; 0-3)          | 97.9                    | 97.9         | 0.79 (0.0001)             | 0.84 (0.0001)            |
| Anxiety                 | 1.40 (1.71; 0-4)            | 0.68 (0.86; 0-3)          | 93.6                    | 100          | 0.71 (0.0004)             | 0.51 (0.02)              |
| Euphoria                | 0.13 (0.52; 0-3)            | 0.08 (0.27; 0-1)          | 100                     | 100          | †                         | †                        |
| Apathy                  | 2.83 (1.55; 0-4)            | 1.35 (0.83; 0-3)          | 97.9                    | 89.4         | 0.74 (0.0002)             | 0.68 (0.001)             |
| Disinhibition           | 0.65 (1.23; 0-4)            | 0.35 (0.66; 0-2)          | 100                     | 100          | 0.97 (0.0001)             | 0.87 (0.0001)            |
| Irritability/Lability   | 1.23 (1.58; 0-4)            | 0.58 (0.81; 0-3)          | 95.7                    | 97.9         | 0.53 (0.02)               | 0.51 (0.02)              |
| Aberrant motor behavior | 1.26 (1.8; 0-4)             | 0.41 (0.64; 0-2)          | 100                     | 97.9         | 0.78 (0.0001)             | 0.77 (0.0001)            |

\* Range refers to the range of behaviors reported by caregivers included in this study.  
† No patient in the test-retest sample was judged to be euphoric, and the reliability of this item could not be assessed.

**Table 3. Correlation of Neuropsychiatric Inventory (NPI) scores and scores of the BEHAVE-AD and the Hamilton Depression Rating Scale (HDRS)**

|                                                       | NPI correlations ( <i>p</i> ) |               |                                 |
|-------------------------------------------------------|-------------------------------|---------------|---------------------------------|
|                                                       | Frequency                     | Severity      | Product of severity × frequency |
| BEHAVE-AD                                             |                               |               |                                 |
| Delusions with NPI delusions                          | 0.75 (0.01)                   | 0.80 (0.01)   | 0.76 (0.01)                     |
| Hallucinations with NPI hallucinations                | 0.74 (0.01)                   | 0.63 (0.01)   | 0.74 (0.01)                     |
| Aggression with NPI aggression                        | 0.78 (0.01)                   | 0.56 (0.01)   | 0.66 (0.01)                     |
| Anxiety and phobias with NPI anxiety                  | 0.66 (0.01)                   | 0.60 (0.01)   | 0.60 (0.01)                     |
| Affective disturbance with NPI dysphoria              | 0.54 (0.01)                   | 0.47 (0.01)   | 0.33 (0.05)                     |
| Activity disturbance with NPI aberrant motor behavior | 0.73 (0.01)                   | 0.75 (0.01)   | 0.77 (0.01)                     |
| Global with NPI global                                | 0.66 (0.0001)                 | 0.71 (0.0001) | 0.66 (0.0001)                   |
| HDRS with NPI dysphoria                               | 0.70 (0.01)                   | 0.59 (0.01)   | 0.62 (0.01)                     |

scale frequency, severity, and product of frequency times severity. All correlations reached the 0.05 significance level and all but one reached the 0.01 significance level, indicating that the NPI is not significantly different from the BEHAVE-AD or the HDRS in its quantification of behavioral symptoms. The weakest correlation was between the Affective Disturbance subscale of the BEHAVE-AD and the Dysphoria subscale of the NPI (frequency × severity  $r = 0.33$ ,  $p = 0.05$ ). The relationship between the HDRS total score and the Dysphoria scale of the NPI was better (frequency × severity  $r = 0.62$ ,  $p = 0.01$ ). The correlation between the NPI Dysphoria scale and the HDRS total ( $r = 0.62$ ) was comparable to the relationship between the BEHAVE-AD Affective Disturbance subscale and the HDRS total ( $r = 0.61$ ). Overall, the analysis demonstrates an acceptable level of concurrent validity compared with the standard instruments.

**Internal consistency reliability.** Cronbach's coefficient alpha was calculated to determine the internal consistency (internal consistency reliability) among the items from the 40 NPIs generated during the concurrent validity study. Cronbach's alpha for overall reliability was 0.88. Cronbach's alpha varied between 0.87 and 0.88 for both severity and frequency of individual items, establishing a high level of internal consistency reliability.

**Item independence.** Twenty-two percent of the items showed significant relationships (Pearson correlation coefficients  $>0.01$ ), and these largely reflected the expected patterns of behavior. For example, hallucination severity was correlated with delusion severity, and depression frequency was related to anxiety (frequency and severity) and apathy (frequency and severity). Seventy-eight percent of the items showed no significant relationships, indicating that they are

independent of each other and are assessing different behaviors.

**Frequency and severity.** Differences emerged between frequency and severity in this analysis. Disinhibition frequency and severity were strongly correlated with agitation frequency (0.004 and 0.001, respectively) but not with agitation severity; anxiety frequency and severity were correlated with apathy frequency (0.005 and 0.043, respectively) but not with apathy severity; disinhibition frequency and severity were correlated with apathy frequency (0.019 and 0.034, respectively) but not apathy severity; and disinhibition frequency and severity were correlated with frequency of aberrant motor behaviors (0.027 and 0.018, respectively) but not their severity. Severity and frequency were significantly correlated for all items ( $p = 0.0001$  for all correlations). This indicates that frequency and severity, while closely related, identify different aspects of some behavioral disturbances.

**Between-rater reliability.** Between-rater reliability was determined using a percent-agreement approach. Between-rater agreement was very high (table 2). Overall, this analysis demonstrates that the NPI is highly reliable when used by different examiners. No trends emerged regarding use of the tool by individuals with different types of training, suggesting that it can be reliably used by individuals with different levels of training when they are provided with appropriate instruction for its use.

**Test-retest reliability.** Table 2 presents the test-retest reliability measures for each of the NPI scales and the overall reliability for severity and frequency. Test-retest reliabilities of all scored measures were significantly correlated; overall correlations were 0.79 for frequency ( $p = 0.0001$ ) and 0.86 for severity ( $p = 0.0001$ ). The weakest correlations (all  $p = 0.02$ ) were for agitation frequency, anxiety severity, and irritability severity. The lower test-retest reliability of these measures may reflect more variability in these behaviors or it may be more difficult for spouses to reliably assess and report these behaviors in response to NPI questions. The second interview was conducted over the telephone and elicited information indistinguishable from that of the personal interview. This suggests that the NPI can be used as a telephone interview instrument, but in this study, the telephone interview always followed the face-to-face interview and the results may have been affected by the earlier encounter.

**Relationship of item scores to dementia severity and age.** Four of the 10 items (frequency × severity) of the NPI had significant relationships with the MMSE: delusions ( $r = -0.31$ ,  $p = 0.05$ ), anxiety ( $r = -0.32$ ,  $p = 0.05$ ), disinhibition ( $r = -0.31$ ,  $p = 0.05$ ), and aberrant motor behavior ( $r = -0.39$ ,  $p = 0.05$ ). These behaviors received higher scores in patients with the lower MMSE scores. Apathy was the only NPI item that showed a significant relationship with age; older patients were rated as more apathetic ( $r = 0.38$ ,  $p = 0.05$ ).

**Normal control subjects.** Forty normal control subjects participated in the study. The maximum possi-

ble score (frequency  $\times$  severity) of any NPI subscale is twelve. In the normal control sample, all NPI scores were zero except for depression, disinhibition, and irritability. Mean depression score (severity  $\times$  frequency) was 0.25 (range, 0 to 6); mean disinhibition score was 0.13 (range, 0 to 4); and mean irritability score was 0.05 (range, 0 to 2). None of the subscale scores was significantly correlated with the MMSE score. Thus, spouses of normal control subjects did not endorse delusions, hallucinations, agitation, anxiety, euphoria, apathy, or aberrant motor behaviors as characteristic behaviors of their partners. Depression, disinhibition, and irritability were identified at low rates among the normal controls.

In the course of examining the normal control subjects, two individuals had MMSE scores in the abnormal range (24 or less), and both had elevated scores on NPI subscales compared with cognitively intact individuals (these two patients were not included among the 40 controls analyzed above). This observation suggests that the NPI may identify deviations from normal behavior in the elderly when cognitive impairments are mild and the individual has not yet exhibited sufficient intellectual decline to mandate clinical assessment.

**Interpretation.** The NPI provides scores for 10 subscales. Each scale is scored for frequency and severity, and a total subscale score can be calculated by multiplying frequency and severity. Elevation of the subscale score beyond that noted among the controls indicates that behavior changes in that domain are present. We did not establish cutoff scores or determine what score might correlate with a clinical diagnosis such as major depressive episode (Dysphoria scale), manic episode (Euphoria scale), or anxiety disorder (Anxiety scale). Categorical diagnoses such as these are made on the basis of clinical information complementary to that obtained with the NPI. The presence of any delusions or hallucinations is abnormal, and depression scores above 6, disinhibition scores above 4, and irritability scores above 2 were not present among normal persons. Scores on the other scales—agitation, euphoria, apathy, and aberrant motor behavior—were not found among normal subjects and, when present, can be regarded as evidence of an abnormality. A global score for the NPI can be generated by summing the total scores (frequency multiplied by severity) of the individual subscales. This global score sums many disparate behaviors (eg, psychosis and apathy) and can be regarded only as an imprecise guide to the total behavioral disturbance exhibited by the patient.

**Discussion.** This study establishes the validity and reliability of the NPI, a new instrument for the assessment of psychopathology in patients with dementia. The content validity study employing a Delphi panel demonstrated that the NPI effectively captures a wide range of psychopathology, and the concurrent validity study showed that the NPI quantitates psychopathology as well as standard instruments such as the BEHAVE-AD and the HDRS. An item analysis

indicated that there is high internal consistency item reliability and that the behavioral domains assessed are largely independent of each other. Between-rater reliability was high, and the test-retest reliability study demonstrated that responses to NPI questions are stable over short intervals (up to 3 weeks). The screening questions were shown to have a less than 5% false-negative rate for detecting psychopathology that would be discovered by completing the entire interview. The results establish the NPI as an instrument in which the clinician can have confidence as a means of assessing and quantitating behavioral changes in dementia.

The NPI has several advantageous features when compared with existing instruments. The screening-question approach allows the clinician to focus on areas of psychopathology while not expending time on unrewarding avenues of inquiry. The caregiver of a patient with little psychopathology is asked only the screening questions in the 10 behavioral domains, and the interview can be completed in 7 to 10 minutes. Caregivers of patients with more psychopathology will require longer interviews.

The NPI explores a wide range of psychopathology. In addition to delusions, hallucinations, depression, anxiety, and agitation that are assessed with instruments currently in use, the NPI also assesses apathy, irritability/lability, disinhibition, and euphoria. Routine investigation of these behavioral domains will allow more comprehensive understanding of the behavioral changes present in any specific dementia patient and may help to distinguish among different types of dementias. In this study assessing a population comprised largely of AD patients, apathy and irritability were among the most commonly recorded behaviors.

The NPI is scored on the basis of information provided by the caregiver and thus avoids the problems inherent in observer-based strategies of asking questions of demented patients or observing behavior for only restricted periods of time. The dysphoria section of the NPI does not include neurovegetative or somatic items and thus avoids the possible confounding effects of the overlapping features of dementia and depression. The NPI distinguishes between behavior severity and frequency, a feature potentially useful in assessing interventions that may affect only one of these behavioral dimensions.

Although we subjected the NPI to comprehensive studies, several areas of application remain unexplored. First, most of the patients in the current studies had AD, and the differential diagnostic utility of the instrument is not established. Second, some behaviors (eg, euphoria) were uncommon in the current study population, and the NPI's ability to assess these behaviors has not been fully determined. Third, the NPI has not been used in treatment studies, and its utility in monitoring behavioral changes related to psychotropic drugs or behavioral interventions remains to be determined. Fourth, the NPI is based on information gathered from caregivers and is subject to bias if the caregiver lacks or distorts information.



With these provisos, the NPI is a useful, psychometrically sound, and efficient means of assessing a wide range of neuropsychiatric alterations in dementia patients.

**Note.** The Neuropsychiatric Inventory described in this report will be provided on request.

## Acknowledgments

The authors thank George Brown and the residents of Leisure World who participated as normal control subjects in this study. Dan Kaufer, Julie Davis, Dianne Christine, Ann Craig, Donna Masterman, Susan McPherson, and Tara Holtzmueller helped to evaluate the patients. The authors are particularly grateful to the following Delphi panel members who aided in the evaluation of the content validity of the Neuropsychiatric Inventory: C. Edward Coffey, MD, Professor of Psychiatry, Allegheny Neuropsychiatric Institute; the late Steven Huber, PhD, Associate Professor of Neurology (Neuropsychology), University of Kansas; William Jagust, MD, Associate Professor of Neurology, University of California-Berkeley; Dillip Jeste, MD, Professor of Psychiatry, Director of Geriatric Psychiatry and Neuropsychiatry, University of California-San Diego; Harvey Levin, PhD, Professor of Psychiatry (Neuropsychology), University of Texas at Galveston; Kim Meader, MD, Associate Professor of Neurology, University of Georgia (Augusta); Mario Mendez, MD, Associate Professor of Neurology, UCLA School of Medicine; William Reichman, MD, Assistant Professor of Psychiatry, Director, Geriatric Psychiatry, Robert Wood Johnson School of Medicine; and Wilfred Van Gorp, PhD, Associate Professor of Psychiatry (Neuropsychology), UCLA School of Medicine.

## References

- Cummings JL, Victoroff JL. Noncognitive neuropsychiatric syndromes in Alzheimer's disease. *Neuropsychiatry Neuropsychol Behav Neurol* 1990;3:140-158.
- Reisberg B, Borenstein J, Salob SP, et al. Behavioral symptoms in Alzheimer's disease: phenomenology and treatment. *J Clin Psychiatry* 1987;48(suppl 5):9-15.
- Burns A, Jacoby R, Levy R. Psychiatric phenomena in Alzheimer's disease. I: Disorders of thought content. *Br J Psychiatry* 1990;157:72-76.
- Burns A, Jacoby R, Levy R. Psychiatric phenomena in Alzheimer's disease. II: Disorders of perception. *Br J Psychiatry* 1990;157:76-81.
- Burns A, Jacoby R, Levy R. Psychiatric phenomena in Alzheimer's disease. III: Disorders of mood. *Br J Psychiatry* 1990;157:81-86.
- Burns A, Jacoby R, Levy R. Psychiatric phenomena in Alzheimer's disease. IV: Disorders of behaviour. *Br J Psychiatry* 1990;157:86-94.
- Cummings JL, Benson DF. *Dementia: a clinical approach*. 2nd ed. Boston: Butterworths-Heinemann, 1992.
- Miller BL, Cummings JL, Villanueva-Meyer J, et al. Frontal lobe degeneration: clinical, neuropsychological, and SPECT characteristics. *Neurology* 1991;41:1374-1382.
- Cummings JL, Miller B, Hill MA, Neshkes R. Neuropsychiatric aspects of multi-infarct dementia and dementia of the Alzheimer type. *Arch Neurol* 1987;44:389-393.
- Dian L, Cummings JL, Petry S, Hill MA. Personality alterations in multi-infarct dementia. *Psychosomatics* 1990;31:415-419.
- Lesser IM, Miller BL, Boone KB, et al. Psychosis as the first manifestation of degenerative dementia. *Bull Clin Neurosci* 1989;54:59-63.
- Petry S, Cummings JL, Hill MA, Shapira J. Personality alterations in dementia of the Alzheimer type. *Arch Neurol* 1988;45:1187-1190.
- Rubin EH, Kinschler DA. Psychopathology of very mild dementia of the Alzheimer type. *Am J Psychiatry* 1989;146:1017-1021.
- Petry S, Cummings JL, Hill MA, Shapira J. Personality alterations in dementia of the Alzheimer type: a three-year follow-up study. *J Geriatr Psychiatry Neurol* 1989;4:203-207.
- Deutsch LH, Bylsma FW, Rovner BW, et al. Psychosis and physical aggression in probable Alzheimer's disease. *Am J Psychiatry* 1991;148:1159-1163.
- Morris RK, Rovner BW, Folstein MF, et al. Delusions in newly admitted residents of nursing homes. *Am J Psychiatry* 1990;147:299-302.
- Steele C, Rovner B, Chase GA, et al. Psychiatric symptoms and nursing home placement of patients with Alzheimer's disease. *Am J Psychiatry* 1990;147:1049-1051.
- Drevets WC, Rubin EH. Psychotic symptoms and the longitudinal course of senile dementia of the Alzheimer type. *Biol Psychiatry* 1989;25:39-48.
- Lopez OL, Becker JT, Brenner RP, Rosen J, Bajulaiye OI, Reynolds CF. Alzheimer's disease with delusions and hallucinations: neuropsychological and electroencephalographic correlates. *Neurology* 1991;41:906-912.
- Mortimer JA, Ebbitt B, Jun S-P, Finch MD. Predictors of cognitive and functional progression in patients with probable Alzheimer's disease. *Neurology* 1992;42:1689-1696.
- Rosen J, Zubenko GS. Emergence of psychosis and depression in the longitudinal evaluation of Alzheimer's disease. *Biol Psychiatry* 1991;29:224-232.
- Stern Y, Mayeux R, Sano M, Hauser WA, Bush T. Predictors of disease course in patients with probable Alzheimer's disease. *Neurology* 1987;37:1649-1653.
- Devanand DP, Sachdev HS, Brown RP, et al. A pilot study of haloperidol treatment of psychosis and behavioral disturbance in Alzheimer's disease. *Arch Neurol* 1989;46:854-857.
- Hamilton M. Development of a rating scale for primary depressive illness. *Br J Soc Clin Psychol* 1967;6:278-296.
- Levin HS, High WM, Goethe KE, et al. The neurobehavior rating scale: assessment of the sequelae of head injury by the clinician. *J Neurol Neurosurg Psychiatry* 1987;50:183-193.
- Devanand DP, Miller L, Richards M, et al. The Columbia University scale for psychopathology in Alzheimer's disease. *Arch Neurol* 1992;49:371-376.
- Alexopoulos GS, Abrams RC, Young RC, Shamoian CA. Cornell scale for depression in dementia. *Biol Psychiatry* 1988;23:271-284.
- Sultzer DL, Levin HS, Mahler ME, et al. Assessment of cognitive, psychiatric, and behavioral disturbances in patients with dementia: the Neurobehavior Rating Scale. *J Am Geriatr Soc* 1992;40:549-555.
- Williams JBW. A structured interview guide for the Hamilton Depression Rating Scale. *Arch Gen Psychiatry* 1988;45:742-747.
- Gottlieb GL, Gur RE, Gur RC. Reliability of psychiatric scales in patients with dementia of the Alzheimer type. *Am J Psychiatry* 1988;145:857-860.
- Cummings JL, Duchen LW. Kluver-Bucy syndrome in Pick disease: clinical and pathologic correlations. *Neurology* 1981;31:1415-1422.
- Cummings JL, Petry S, Dian L, et al. Organic personality disorder in dementia syndromes: an inventory approach. *J Neuropsychiatry Clin Neurosci* 1990;2:261-267.
- Flynn FG, Cummings JL, Gornbein J. Delusions in dementia syndromes: investigation of behavioral and neuropsychological correlates. *J Neuropsychiatry Clin Neurosci* 1991;3:364-370.
- Folstein MF, Folstein SE, McHugh PR. The "mini-mental state": a practical method for grading the cognitive state of patients for the clinician. *J Psychiatr Res* 1975;12:189-198.
- Cronbach LJ. Coefficient alpha and the internal structure of tests. *Psychometrika* 1951;16:294-334.
- McKhann G, Drachman D, Folstein M, Katzman R, Price D, Stadlan EM. Clinical diagnosis of Alzheimer's disease: report of the NINCDS-ADRDA Work Group under the auspices of Department of Health and Human Services Task Force on Alzheimer's Disease. *Neurology* 1984;34:939-944.
- Román GC, Tatemichi TK, Erkinjuntti T, et al. Vascular dementia: diagnostic criteria for research studies. Report of the NINDS-AIREN International Workshop. *Neurology* 1993;43:250-260.

# Neurology®

## **The Neuropsychiatric Inventory: Comprehensive assessment of psychopathology in dementia**

J. L. Cummings, M. Mega, K. Gray, et al.

*Neurology* 1994;44;2308

DOI 10.1212/WNL.44.12.2308

**This information is current as of December 1, 1994**

### **Updated Information & Services**

including high resolution figures, can be found at:  
<http://www.neurology.org/content/44/12/2308.full.html>

### **Citations**

This article has been cited by 100 HighWire-hosted articles:  
<http://www.neurology.org/content/44/12/2308.full.html##otherarticles>

### **Permissions & Licensing**

Information about reproducing this article in parts (figures, tables) or in its entirety can be found online at:  
<http://www.neurology.org/misc/about.xhtml#permissions>

### **Reprints**

Information about ordering reprints can be found online:  
<http://www.neurology.org/misc/addir.xhtml#reprintsus>

*Neurology*® is the official journal of the American Academy of Neurology. Published continuously since 1951, it is now a weekly with 48 issues per year. Copyright © 1994 by the American Academy of Neurology. All rights reserved. Print ISSN: 0028-3878. Online ISSN: 1526-632X.

